Home Reading 01 — Introduction

Before Class

Preview Lecture 1

Read Lecture 1: Introduction.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 1-2, pp. 3–5 (mathematical modelling).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

Dynamic models, model boundaries, inputs, outputs, states, model order and first-principles balance equations.

Tutorial Preparation

Read Tutorial 1 before class. Be ready to:

- Q1: Identify the inputs, outputs, states and model order.
- Q2: Reduce the spring assembly, use the supplied effective mass, and find initial acceleration and final displacement.
- Q3: Identify inputs and states from the supplied thermal balances, then calculate the instantaneous temperature rates.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why do modelling assumptions determine what a model can predict?
- How can changing the model boundary change the number of independent states?

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